

Greenhouse Controller Evolution Notes (Python Refactor Summary)

Overview

This document summarizes the evolution of a Raspberry Pi greenhouse control system from legacy Python 2 scripts to a modern Python 3 architecture with database-driven control, sensor validation, and safety improvements.

Key System Components

1. checksensors Script

- DS18B20 multi-sample reading (5 samples) - Outlier filtering (-50 to 100°C, remove 85.0) - Trimmed averaging for stability - Writes to MySQL currenttemp table - Timestamp-based freshness tracking

2. Controller v4.1

- DB-driven greenhouse automation controller - Hysteresis control for heater, fans, windows - Override system (fan + window) - Sensor fallback hierarchy (Average → Front → Back) - Stale sensor detection (>60s) - Preserved original window motion timings

3. Data Flow Architecture

```
graph TD
    A[DS18B20 Sensors] --> B[checksensors (sampling + validation)]
    B --> C[MySQL currenttemp table]
    C --> D[Controller v4.1 decision engine]
    D --> E[GPIO outputs (heater, fans, windows)]
```

Design Decisions

- Separation of sensing and control layers - Database as integration buffer - Safety-first fallback behavior - Mechanical protection via preserved timing sequences - Cron-based execution model ensures resilience

Safety Mechanisms

- Emergency shutdown on invalid sensor state - GPIO cleanup on exit - Sensor staleness detection - Input validation and retry logic

Future Improvements

- Sensor health scoring system - Predictive failure detection engine - Unified config/threshold table - systemd migration option